Correlation-Aware Raw-Flow Fusion in IMPACT for GNSS-Denied UAV Navigation

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***Abstract*—Downward optical flow and a LiDAR-inertial-visual estimator can depend on the same inertial measurements and range-derived geometry. Fusing a derived flow velocity as an independent measurement can therefore omit relevant error correlations. We present a research extension of FAST-LIVO2 within the IMPACT navigation framework. A 28-dimensional error state augments the native 19-dimensional state with a cloned pose and an uncertain local plane anchored by one simultaneous noisy range observation. Raw two-view pixels are fused without gyro precompensation, and native inertial, LiDAR and visual updates propagate or condition the joint covariance. A tagged gyro covariance component enables targeted correlation ablations. Original-acquisition replay separates sensor publication, estimator processing and independent truth evaluation. On one recorded simulated UAV flight, the full model applies 37 nonzero updates to the actual native state and covariance; independent reconstruction verifies the projection derivatives, gains, injections and Joseph covariance updates. The matched joint/no-flow control yields a trajectory RMS error of 7.20 mm, compared with 6.64 mm for the full model on this sequence. The present evidence establishes integration and numerical reproducibility under a local-plane, fixed-map approximation. It does not establish calibrated integrity bounds, general statistical superiority or hardware readiness.**

***Index Terms*—multisensor fusion, optical flow, LiDAR-inertial-visual odometry, stochastic cloning, UAV navigation**

I. Introduction

GNSS-denied UAV navigation combines measurements with different geometric strengths and failure modes. A forward camera supplies appearance constraints, a LiDAR supplies spatial geometry, and an IMU connects successive states. A downward camera and a range sensor offer another view of nearby surfaces, especially during low-altitude motion. These sensors are complementary, but their estimation errors need not be independent. A flow velocity obtained using inertial rotation compensation and a measured height already depends on quantities used by the main navigation estimator.

FAST-LIO2 and FAST-LIVO2 provide efficient LiDAR-inertial and LiDAR-inertial-visual estimation backbones, respectively [1], [2]. Integrating another observation into such a backbone requires more than publishing a topic or reducing a covariance. The observation must change the state used by the estimator, retain the intended shared uncertainty, and preserve the acquisition-time relationship between its constituent measurements. A single pencil range is also not a depth measurement for every tracked image pixel.

This paper studies a specific extension of the IMPACT research stack: correlation-aware raw-flow updates using a range-anchored local plane and a stochastic pose clone. The retained plane is a nuisance variable shared by the pixels, rather than a collection of independent per-pixel heights. Two-view reprojection explains rotation through the estimated poses, so the flow frontend does not subtract an additional gyro-derived rotation. The implementation connects to the actual native state and covariance, including conditioning the auxiliary variables when native LiDAR or visual updates occur.

The contributions are: (i) a concrete native-state integration of raw two-view pixels and a noisy range-anchored plane, with analytic derivatives and explicit local validity conditions; (ii) a covariance decomposition that supports separate gyro-cross and plane-cross ablations while retaining positive-semidefinite covariance components; and (iii) a reproducible five-input simulation replay with an independent reconstruction of accepted updates. Stochastic cloning, the EKF, and manifold error resets are established methods; the contribution is their specific integration and the evidence needed to test the stated correlation model. The full mathematical specification is provided in the appendices.

A. Related Work and Scope

MSCKF established efficient use of multiple camera poses in a visual-inertial filter [3]. OpenVINS provides an extensively documented platform for on-manifold filtering and calibration [4]. ROVIO uses direct photometric information in an EKF, while VINS-Mono uses tightly coupled optimization with inertial and feature observations [5], [6]. These approaches motivate explicit treatment of the state, its errors and sensor timing; their published performance is not a baseline result of the present experiment.

FAST-LIVO2 combines direct LiDAR and visual updates on a common map [2]. Subsequent work addresses its memory and computation on resource-constrained platforms [7]; this does not establish performance of our implementation on an Orin NX. Our downward frontend uses established corner selection and pyramidal Lucas–Kanade tracking [8], [9]. The right-perturbation convention and injection reset follow standard error-state filtering [10].

Unlike a complete mapping-consistency analysis, this extension treats the native map as fixed and native measurement updates as conditionally independent. Historical map–state correlations, cross-feature image noise, and selection-induced covariance effects are not reconstructed. The term "correlation-aware" refers to the retained inertial/clone/plane

covariance within this declared model, not to global statistical consistency.

II. System and Acquisition-Time Contract

The software stack uses ROS 2, a research port of FAST-LIVO2, an independent downward image frontend, and an ArduPilot ExternalNav execution interface. Fig. 1 separates the estimator from the evaluator and from the flight-controller interface. Estimation receives only sensor data; truth and the noise-free range reference are reserved for offline evaluation. The current estimation experiment replays a previously recorded simulated flight. It neither drives a new flight controller nor feeds truth to the estimator.

A. Sensors, Frames and Timing

The recording contains LiDAR snapshots, IMU messages, forward RGB images, downward RGB images and noisy downward range messages. Images are 320×240 pixels with zero distortion and intrinsics $f_x = f_y = 277.128129211$, $c_x = 159.5$, $c_y = 119.5$. The downward optical frame has the declared body transform

$$ {}^{B}R_C = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \qquad {}^{B}t_C = \begin{bmatrix} 0 & 0 & -0.03 \end{bmatrix}^{\top} \text{ m}. \tag{1} $$

The range origin is colocated with this camera and its body ray is $\mathbf{d}_B = (0, 0, -1)^{\top}$. Simulation extrinsics and intrinsics are configuration values, not a calibration of the intended physical sensors. The LiDAR data are instantaneous simulated snapshots with no invented within-scan acquisition times.

Forward and downward image acquisition stamps match exactly. Their observed interval is 66 ms, whereas range samples lie on a 20 ms grid. Exact camera/range coincidences occur every 660 ms. The first profile starts a pair only at such an exact coincidence and ends it at the next original image epoch. Frames are not reused between pairs. Other range samples remain part of the receipt audit but are not fused in this profile. No nearest range is restamped to make it appear simultaneous.

All modes receive the same original serialized sensor records. A cumulative barrier gives the counts published up to an acquisition epoch; processing waits until the corresponding callbacks have actually arrived. Completion requires all five final counts, their acquisition tails, the final barrier, and classification of every forward image. Thus a publisher finishing does not by itself certify estimator completion.

B. Asynchronous LiDAR and Camera Epochs

The cameras run faster than the LiDAR snapshots. At an image epoch without new LiDAR points, IMU propagation still advances the state and the existing map supports the forward visual update. The LiDAR measurement update is skipped, the number of fresh points is recorded as zero, and an old cloud is not presented as a new scan. A real initialized LiDAR map is required before the first published visual state. Propagation starts from an explicit stationary-initialization epoch; empty-cloud deskew history is cleared.

C. Execution Interface and Target Hardware

The firmware target is official ArduCopter 4.7.1, with source commit `dbe79216`. Independent compilation and a disarmed smoke test verify the runtime version and complete parameter table. Generic MAVLink odometry uses `VISO_TYPE=1`; simulated GNSS is disabled using the version's explicit parameter names. ArduPilot documents the external-navigation message interface [11].

Separate unarmed MAVROS/Gazebo trials matched 557 synthetic-source and 585 FAST-LIO-source adapter outputs to ordered wire odometry and FCU logs. These are interface tests, not flights using the new raw-flow state; receipt matching also does not prove EKF fusion. Their detailed scope is reported in the planning/execution appendix.

The planned physical platform comprises a Livox Mid-360, camera, MicoAir downward flow module, NVIDIA Orin NX and CUAV X25 MEGA. Camera and flow-module models are not yet fixed. None of the reported timing or accuracy results is a measurement on that hardware. Acquisition-time validity, MAVROS discovery, FCU receipt health, EKF use and task completion remain distinct conditions; a monotone transport timestamp or a healthy receipt bit cannot replace them.

III. Correlation-Aware Raw-Flow Fusion

State and propagation. The research port retains the native nineteen-state representation and LiDAR/direct-visual measurement models of FAST-LIVO2 [2]. Its right-local error, with zero-based indices, is

$$ \begin{aligned} \delta\mathbf{x} = \big[& \delta\boldsymbol{\theta}^{\mathsf{T}}, \delta\mathbf{p}^{\mathsf{T}}, \delta\tau, \delta\mathbf{v}^{\mathsf{T}}, \\ & \delta\mathbf{b}_g^{\mathsf{T}}, \delta\mathbf{b}_a^{\mathsf{T}}, \delta\mathbf{g}^{\mathsf{T}} \big]^{\mathsf{T}} \in \mathbb{R}^{19}, \end{aligned} \tag{2} $$

where τ is inverse exposure, at index 6; velocity starts at index 7. We augment it with a six-dimensional pose clone and a three-dimensional local plane, giving $\delta\mathbf{y} = [\delta\mathbf{x}^{\mathsf{T}}, \delta\boldsymbol{\theta}_0^{\mathsf{T}}, \delta\mathbf{p}_0^{\mathsf{T}}, \delta\boldsymbol{\pi}^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{28}$. Here $R_{\text{true}} = R\,\mathrm{Exp}(\delta\boldsymbol{\theta})$ and $\boldsymbol{\pi} = (a, b, h)$ represents $z = ax + by + h$ in world coordinates. Native IMU substeps internally accumulate their F_j, Q_j, exporting the aggregate transition and noise per call, including a separate gyro-noise component. The substep joint transition is $\Phi_j = \mathrm{blkdiag}(F_j, I_9)$, so

$$ P_{j+1}^{-} = \Phi_j P_j^{+} \Phi_j^{\mathsf{T}} + \mathrm{blkdiag}(Q_j, 0_9). \tag{3} $$

This includes all retained cross blocks. We retain the implemented native Δt^2 process-noise convention, rather than silently substituting a continuous-time spectral-density model; Appendix A-B specifies it. Pose cloning and error-state filtering themselves are established techniques [3], [10].

Uncertain range-anchored geometry. At a simultaneous camera/range epoch, a measured range ρ along the known body ray $(\mathbf{t}_r, \mathbf{d}_r)$ defines $\mathbf{q} = \mathbf{p} + R(\mathbf{t}_r + \rho\mathbf{d}_r)$. With $\mathbf{n} = (-a, -b, 1)^{\mathsf{T}}$, set $h = \mathbf{n}^{\mathsf{T}}\mathbf{q}$. The slope prior has zero mean and standard deviation 0.05 per axis; the seed range has variance $(0.01\,\mathrm{m})^2$ in this simulation profile. Neither a true ground height nor a truth pose enters the estimator. Linearizing the clone and plane construction gives

$$ P_0 = B\ \mathrm{blkdiag}(P_{xx}, \Sigma_{ab}, \sigma_\rho^2) B^{\mathsf{T}}. \tag{4} $$

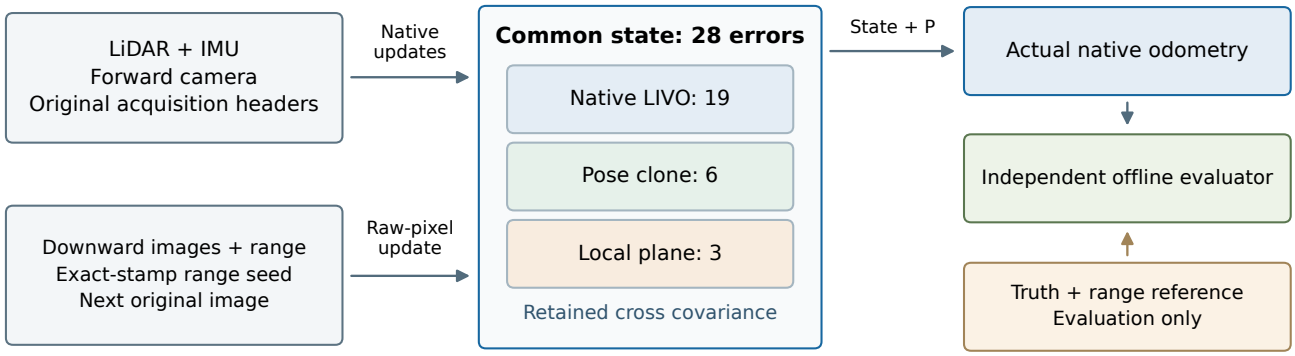


Fig. 1. Implemented estimation path and separate evaluation boundary. The AP 4.7.1 interface is prepared independently; a completed closed-loop flight using the new raw-flow estimator is outside the present evidence.

The complete B is given in Appendix A-C. Thus the plane height, clone, and native pose are correlated, and the initial clone makes P_0 positive semidefinite rather than strictly positive definite. The seed range is consumed once by this construction. The reported replay uses an exact-stamp seed and the next camera image as a nonoverlapping pair; it does not additionally fuse that same range as an independent update.

Raw two-view observation. For a first pixel $\mathbf{u}_0$, let $\boldsymbol{\nu}(\mathbf{u}_0)$ be its unit-optical-Z pinhole ray. Camera extrinsics $(R_{bc}, \mathbf{t}_c)$ map optical coordinates to the IMU body. The ray–plane prediction is

$$\begin{aligned} \mathbf{o}_0 &= \mathbf{p}_0 + R_0\mathbf{t}_c, \quad \mathbf{d}_0 = R_0 R_{bc}\boldsymbol{\nu}(\mathbf{u}_0), \\ \lambda &= \frac{h - \mathbf{n}^\mathsf{T}\mathbf{o}_0}{\mathbf{n}^\mathsf{T}\mathbf{d}_0}, \quad \mathbf{q} = \mathbf{o}_0 + \lambda\mathbf{d}_0, \\ \widehat{\mathbf{u}}_1 &= \mathrm{proj}\big(R_{bc}^\mathsf{T}[R_1^\mathsf{T}(\mathbf{q} - \mathbf{p}_1) - \mathbf{t}_c]\big)\,. \end{aligned} \tag{5}$$

Lucas–Kanade correspondences [9] provide the raw endpoint pixels. No gyro-precompensated flow or derived velocity is fused. Writing $\mathbf{r}_i = \mathbf{u}_{1i} - \widehat{\mathbf{u}}_{1i}$, $H_i = \partial\widehat{\mathbf{u}}_{1i}/\partial\delta\mathbf{y}$, and $U_i = \partial\widehat{\mathbf{u}}_{1i}/\partial\mathbf{u}_{0i}$, the endpoint noise model is

$$\mathcal{R}_i = U_i\Sigma_{0i}U_i^\mathsf{T} + \Sigma_{1i}. \tag{6}$$

The current implementation supplies $\Sigma_{0i} = \Sigma_{1i} = I_2$ in pixel-squared units. This is an explicit engineering covariance, not an empirically calibrated uncertainty model for the tracker. All analytic derivatives are provided in Appendix A-D.

Conditioning on native updates. Native LiDAR and frontal-image updates must also affect the auxiliary state. Partition the prior into native $\mathbf{x}$ and auxiliary $\mathbf{z} \in \mathbb{R}^9$, set $A = P_{xx}^-$ and $D = P_{zx}^- A^{-1}$, and denote the native returned covariance before injection reset by B_x. For its nominal correction $\Delta\mathbf{x}$, use

$$\begin{aligned} \Delta\mathbf{z} &= D\Delta\mathbf{x}, \\ \widetilde{P}_{xx}^+ &= B_x, \quad \widetilde{P}_{xz}^+ = B_x D^\mathsf{T}, \\ \widetilde{P}_{zz}^+ &= P_{zz}^- - D(A - B_x)D^\mathsf{T}. \end{aligned} \tag{7}$$

This is a Gaussian conditional extension under a fixed-map, independent-native-observation approximation. It does not reconstruct historical map–state correlations omitted by the native estimator. Noncontractive or invalid native covariance changes are rejected by the integration contract.

Joint update and injection. Stack the accepted residuals, Jacobians, and block-diagonal endpoint covariances as $(\mathbf{r}, H, \mathcal{R})$. The full update is

$$\begin{aligned} S &= HP^-H^\mathsf{T} + \mathcal{R}, \\ K &= P^-H^\mathsf{T}S^{-1}, \quad \Delta\mathbf{y} = K\mathbf{r}, \quad L = I - KH, \\ \widetilde{P}^+ &= LP^-L^\mathsf{T} + K\mathcal{R}K^\mathsf{T}, \\ P^+ &= \Gamma\widetilde{P}^+\Gamma^\mathsf{T}. \end{aligned} \tag{8}$$

We inject the corrections into the actual native nominal state and the clone/plane. Both right-local rotation errors use the right Jacobian reset [10]. With $J_1 = J_r(\Delta\boldsymbol{\theta}_1)$ and $J_0 = J_r(\Delta\boldsymbol{\theta}_0)$, let $\Gamma = \mathrm{blkdiag}(J_1, I_{16}, J_0, I_6)$. The reset native block is written back to the native covariance. The same reset convention is used after (7). The joint formulation allows a raw pixel residual to affect velocity and bias through covariance even though their direct observation columns are zero.

Targeted ablations and validity. We separately propagate a positive semidefinite component P_g explained by the initial gyro-bias random variable and subsequent native gyro/gyro-bias process noise. Let $\mathcal{D}_g = \mathcal{D}_{19|6|3}$ zero cross blocks between native, clone, and plane groups, and let $\mathcal{D}_\pi = \mathcal{D}_{25|3}$ separate the plane. The two deliberately misspecified controls are

$$\begin{aligned} &\text{gyro: } P' = P - P_g + \mathcal{D}_g(P_g), \quad P_g' = \mathcal{D}_g(P_g), \\ &\text{plane: } P' = \mathcal{D}_\pi(P), \quad P_g' = \mathcal{D}_\pi(P_g). \end{aligned} \tag{9}$$

They preserve group marginals and the decomposition $P - P_g \succeq 0$; they do not zero all cloned-pose correlation. Geometry, feature support, innovation, correction-size, and covariance gates define a bounded local operating domain (Appendix A-I). These are engineering rejection rules, not calibrated integrity bounds. The reported study establishes actual integration and reproducibility within these assumptions; a universal observability or consistency guarantee is not asserted.

IV. Recorded-Flight Experiments

A. Setup and Matched Controls

We evaluate one previously recorded simulated UAV flight, with an original acquisition interval of 83.753 s. The five input streams contain 16 586 IMU messages, 830 instantaneous LiDAR snapshots, 1270 forward images, 1270 downward images and 4188 noisy range samples. The independent truth record shows approximately 2.08 m maximum horizontal displacement from the initial position and a 2.07 m vertical span. Thus the record includes translation and altitude change, rather than only stationary filter drift. Truth and the noise-free range reference are never estimator inputs.

Every mode consumes the same serialized sensor records, acquisition headers and cumulative barriers, using the same executable and build manifest. An independent audit verifies every published source-row identifier and payload hash against the common sensor index. Each mode receives all five streams and completes 1253 post-visual-update image epochs. All 1270 forward images are classified: 14 startup discards, two IMU-initialization epochs, 1253 processed epochs and one terminal epoch outside the available joint LiDAR/IMU coverage. These classifications are retained rather than silently trimming the input. Replay runs at half acquisition speed; this is a controlled software experiment, not an Orin NX runtime benchmark.

We compare five modes. *Native* uses the native estimator's original reset handling. *Joint/no-flow*, the primary flow control, uses the same auxiliary-state propagation and manifold resets as the full method but disables raw-flow corrections. *Full* retains the joint covariance. *Zero gyro cross* removes only cross-block terms of the tagged gyro covariance contribution, while retaining the other covariance components. *Zero plane cross* removes the plane-to-other-state cross blocks. The two ablations retain their marginal variances and are checked for positive semidefiniteness. They deliberately change the correlation model; they are not independently calibrated estimators.

The declared engineering settings are a one-pixel standard deviation at each tracked image endpoint, 0.01 m range standard deviation, and 0.05 prior standard deviation for each local plane slope. The pixel covariance is not a measured camera-noise calibration. Exact acquisition coincidences provide 127 potential range-seeded pairs; startup and eligibility checks leave 123 completed nonoverlapping pairs. This low-duty profile does not fuse every range sample.

B. Independent Evaluation

After the estimator processes exit, a separate evaluator associates each odometry epoch with the nearest original truth epoch within 15 ms. All five modes have 1253 matches, no unmatched estimates and a maximum observed association offset of 9 ms. One initial SE(3) transform fixes the coordinate gauge; neither scale fitting nor a trajectory-wide alignment is used. For aligned truth position $\mathbf{p}_i^*$ and estimate $\hat{\mathbf{p}}_i$, the reported errors are

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|\hat{\mathbf{p}}_i - \mathbf{p}_i^*\|^2}, \qquad E_{\max} = \max_i \|\hat{\mathbf{p}}_i - \mathbf{p}_i^*\|. \tag{10}$$

The $N = 1253$ temporally related estimates are samples of one flight, not 1253 independent trials.

For every accepted batch, the evaluator reconstructs the raw-pixel residual, finite-difference 28-state Jacobian, two-endpoint image-noise mapping, innovation covariance, gain, state injection and Joseph update with both SO(3) resets. It verifies the resulting native 19-state mean and covariance, including the identity of the native state object used by the LiDAR and forward-visual updates. It also reconstructs the two declared ablation masks. This check uses recorded inputs and matrices; it does not infer successful fusion from an odometry publication count.

TABLE I
Matched-input results on one simulated flight. ATE uses one initial SE(3) gauge alignment, without scale fitting. Each row has 1253 matched estimates.

Method	Updates	RMS (mm)	Max (mm)	NIS/DOF†
Native	0	7.369	121.779	–
Joint/no-flow	0	7.203	117.431	–
Full	37	6.645	116.141	0.024
Zero gyro cross	37	6.768	117.981	0.025
Zero plane cross	37	6.742	117.396	0.026

†Median over accepted flow updates; not a calibration test.

C. Results and Scope

Table I and Fig. 2 report all five modes. The full model has 6.645 mm RMS and 116.141 mm maximum position error. Its RMS is 0.558 mm below the matched joint/no-flow control in this record. The gyro-cross and plane-cross ablations have RMS errors of 6.768 and 6.742 mm, respectively. Their small differences show that the retained correlations affect the result, but a single flight cannot establish their general accuracy benefit. The difference from the native mode also includes changed reset handling and cannot be attributed entirely to optical flow.

The full method applies 37 nonzero corrections to both the actual native state and covariance. Of the 123 evaluated pairs, 47 are rejected because predicted translation is below the 5 mm threshold and 39 because no tracks survive the frontend checks. Both ablations also apply 37 corrections. For the full method, the largest absolute Frobenius discrepancy between the recorded projection Jacobian and central finite differences is 4.45×10^{-7}. All accepted updates pass the independent gain, injection and covariance reconstruction checks. The median accepted NIS per degree of freedom is 0.024; residual selection and the engineering noise settings prevent interpreting this value as evidence of calibrated consistency.

These results establish actual nonlinear fusion and numerical reproducibility within the declared local-plane and

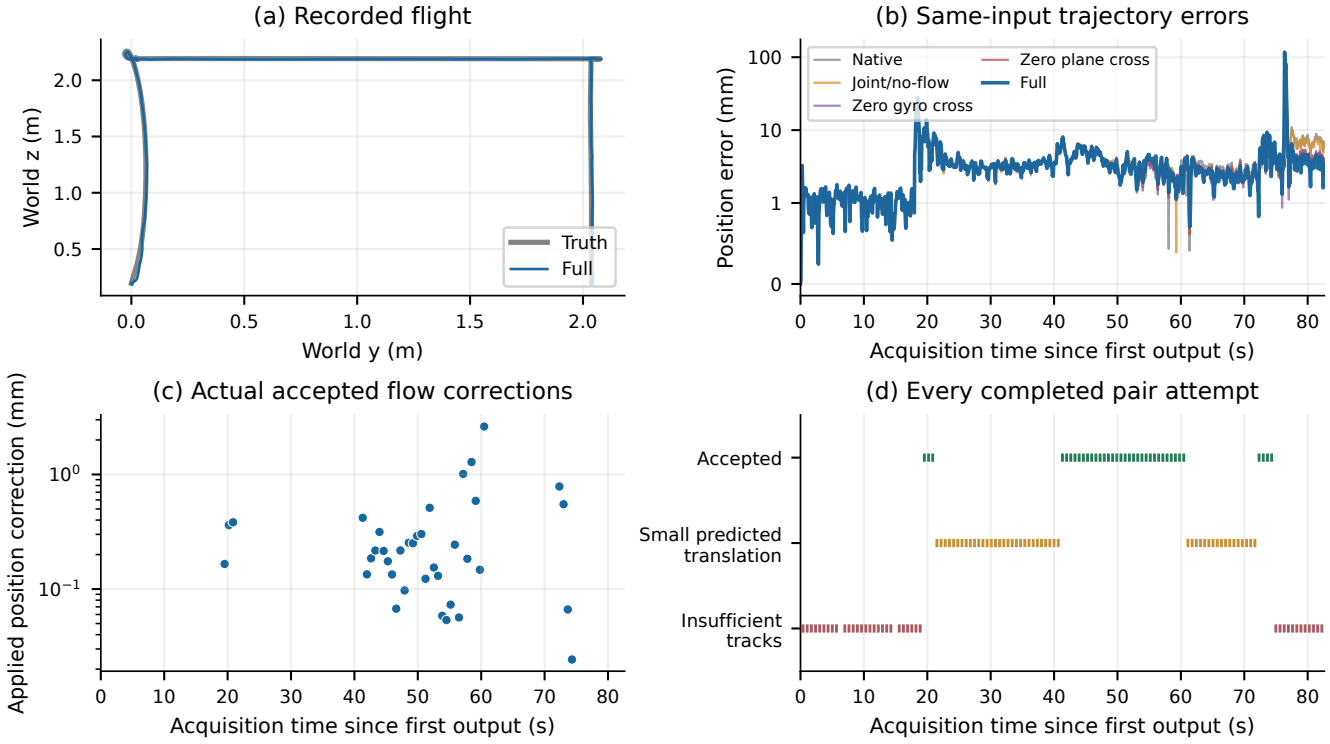


Fig. 2. Five modes evaluated on the same recorded simulated flight. The trajectory and error curves use one initial SE(3) gauge alignment without scale fitting. Summary errors describe this record only; they do not establish a statistical performance ranking.

fixed-map model. They do not establish multiseed statistical superiority, robustness to nonplanar terrain or low texture, calibrated integrity bounds, hardware timing, or a complete ArduCopter 4.7.1 mission using this estimator. Those evaluations remain future work. Deterministic dropout-selection tools are available, but no unexecuted degradation matrix is included as a result here.

V. Discussion and Limitations

The experiment verifies that downward raw pixels can update the same state used by the native LiDAR and forward-visual estimator while retaining the declared clone, plane and gyro covariance components. The independent reconstruction is stronger evidence of this integration than a topic publication counter. Nevertheless, positive-semidefinite covariance, small trajectory error, or accepted NIS values do not establish a calibrated probability of navigation failure.

The local plane is a finite-domain approximation. Surface discontinuities, nonplanar ground, insufficient image texture and long gaps require further evaluation. Pair selection prevents direct reuse of an image endpoint in successive flow pairs, but it does not prove statistical independence of image features or restore map correlations. The fixed one-pixel endpoint standard deviation is an engineering setting; innovation selection also changes the distribution of accepted residuals. NIS values are therefore reported as diagnostics, not a calibration certificate.

The principal no-flow control shares the new covariance resets and auxiliary-state handling. An additional native control exposes the effect of using the original reset handling, so improvements cannot automatically be attributed to flow. A single recorded input supplies no independent multiseed evidence. Strong serial-estimation comparisons, calibrated pixel/range uncertainty, sensor-dropout and nonplanar scenarios, a complete AP 4.7.1 mission using this estimator, and physical-platform measurements remain necessary before general performance or deployment claims.

IMPACT's existing integrity-margin planning layer is described separately in the appendix. Its historical simulation evidence uses a different estimation chain. The new joint covariance must not be substituted into an old protection-level calibration and relabeled a validated bound. Task-level risk and perception-budget claims require a new joint evaluation.

VI. Conclusion

We implemented a 28-dimensional raw-flow extension of a LiDAR-inertial-visual estimator with a noisy range-anchored plane and a pose clone. Acquisition-time pairing, retained cross-covariances, manifold resets and targeted ablations make its assumptions testable. Native-state corrections and independent reconstruction establish numerical reproducibility on one simulation input. Broader accuracy, integrity and deployment claims require the additional experiments identified above.

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APPENDIX A
IMPLEMENTED ESTIMATOR: COMPLETE LOCAL DERIVATIONS

This appendix specifies the implemented research extension, including its coordinate conventions, numerical approximations, and rejection conditions. It is intended to make the executable model reconstructible. Standard stochastic cloning, Gaussian conditioning, and error-state Kalman filtering are not claimed as original contributions [3], [10]. The native LiDAR and frontal-image estimator is FAST-LIVO2 [2]; the extension does not replace its map construction or direct-image frontend.

A. Coordinates, Error State, and Injection

Let $R \in \mathrm{SO}(3)$ map IMU/body coordinates to world coordinates. Positions, velocities, and gravity are expressed in the world; inertial biases are in body coordinates. A camera point $\mathbf{c}$ is mapped into the body by $R_{bc}\mathbf{c} + \mathbf{t}_c$. All positions are in metres, rotations in radians, image coordinates in pixels, and acquisition times in seconds. The native error ordering is

$$\delta\mathbf{x} = \mathrm{col}\left(\underbrace{\delta\boldsymbol{\theta}}_{0:2}, \underbrace{\delta\mathbf{p}}_{3:5}, \underbrace{\delta\tau}_{6}, \underbrace{\delta\mathbf{v}}_{7:9}, \underbrace{\delta\mathbf{b}_g}_{10:12}, \underbrace{\delta\mathbf{b}_a}_{13:15}, \underbrace{\delta\mathbf{g}}_{16:18}\right). \tag{11}$$

Indices in this appendix are zero-based. The scalar τ is inverse exposure, not velocity, range, or a time offset. Nominal injection is

$$\begin{aligned} &R^+ = R^- \operatorname{Exp}(\Delta\boldsymbol{\theta}), \quad \mathbf{p}^+ = \mathbf{p}^- + \Delta\mathbf{p}, \quad \tau^+ = \tau^- + \Delta\tau, \\ &\mathbf{v}^+ = \mathbf{v}^- + \Delta\mathbf{v}, \quad \mathbf{b}_g^+ = \mathbf{b}_g^- + \Delta\mathbf{b}_g, \quad \mathbf{b}_a^+ = \mathbf{b}_a^- + \Delta\mathbf{b}_a, \\ &\mathbf{g}^+ = \mathbf{g}^- + \Delta\mathbf{g}. \end{aligned} \tag{12}$$

The difference between native states before and after a returned update uses $\operatorname{Log}((R^-)^{\mathsf{T}}R^+)$ and the corresponding additive differences, in the ordering (11). The joint error is

$$\delta\mathbf{y} = [\delta\mathbf{x}^{\mathsf{T}}, \underbrace{\delta\boldsymbol{\theta}_0^{\mathsf{T}}, \delta\mathbf{p}_0^{\mathsf{T}}}_{\text{clone, indices 19:24}}, \underbrace{\delta a, \delta b, \delta h}_{\text{plane, indices 25:27}}]^{\mathsf{T}}. \tag{13}$$

Both pose rotations use right-local errors. The plane is an additive world-coordinate chart, not a unit-normal rotation chart:

$$\mathbf{n} = (-a, -b, 1)^{\mathsf{T}}, \qquad \mathbf{n}^{\mathsf{T}}\mathbf{q} = h. \tag{14}$$

Its normal is deliberately not normalized in the intersection equations.

B. Actual Native IMU Propagation and Noise Convention

For each native subinterval Δt, let the midpoint raw inertial samples be $\overline{\boldsymbol{\omega}}_m$ and $\overline{\mathbf{a}}_m$. The implementation uses

$$\boldsymbol{\omega} = \overline{\boldsymbol{\omega}}_m - \mathbf{b}_g, \qquad \mathbf{a} = \alpha\overline{\mathbf{a}}_m - \mathbf{b}_a, \qquad \alpha = 9.81/\|\overline{\mathbf{a}}_{\mathrm{init}}\|. \tag{15}$$

The scale α is inherited from the native initialization, rather than being estimated as an additional research state. With R denoting the pre-substep rotation, the native nominal equations are

$$\begin{aligned} &R' = R\operatorname{Exp}(\boldsymbol{\omega}\Delta t), \qquad \mathbf{a}_w = R'\mathbf{a} + \mathbf{g}, \\ &\mathbf{p}' = \mathbf{p} + \mathbf{v}\Delta t + \tfrac{1}{2}\mathbf{a}_w\Delta t^2, \quad \mathbf{v}' = \mathbf{v} + \mathbf{a}_w\Delta t, \end{aligned} \tag{16}$$

with nominal biases, gravity, and inverse exposure unchanged by this propagation step. In particular, the acceleration uses the updated R'. The covariance implementation is the native first-order discrete model below; it is not replaced by the exact Jacobian of every term in (16).

Let $c_b, c_g, c_\tau \in \{0, 1\}$ denote enabled bias, gravity, and exposure estimation respectively, and let $E = \operatorname{Exp}(-\boldsymbol{\omega}\Delta t)$. In the seven block groups of (11),

$$F = \begin{bmatrix} E & 0 & 0 & 0 & -c_b I_3\Delta t & 0 & 0 \\ 0 & I_3 & 0 & I_3\Delta t & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -R[\mathbf{a}]_\times\Delta t & 0 & 0 & I_3 & 0 & -c_b R\Delta t & c_g I_3\Delta t \\ 0 & 0 & 0 & 0 & I_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I_3 \end{bmatrix}. \tag{17}$$

There are no implemented $\mathbf{p}$–attitude, $\mathbf{p}$–accelerometer-bias, or $\mathbf{p}$–gravity Δt^2 blocks in F. Including them in a paper formula while leaving the executable unchanged would describe a different estimator. Let $D_\omega, D_a, D_{b_g}, D_{b_a}$ be the diagonal matrices

populated from the native covariance-coefficient vectors, and q_τ the corresponding exposure coefficient. The actual discrete noise is

$$Q = \Delta t^2 \,\mathrm{blkdiag}\left(D_\omega, 0_3, c_\tau q_\tau, R D_a R^\mathsf{T}, D_{b_g}, D_{b_a}, 0_3\right). \tag{18}$$

These entries are used as stored covariance coefficients; they are not squared again. Equation (18) is a retained per-step Δt^2 convention, not a claim that the coefficients are calibrated continuous-time noise densities. Process cross terms absent from the native code are absent here as well.

For substeps $j = 1, \ldots, M$ within a native call, initialize $\overline{F}_0 = I_{19}$ and $\overline{Q}_0 = 0$, then accumulate

$$\overline{F}_j = F_j \overline{F}_{j-1}, \qquad \overline{Q}_j = F_j \overline{Q}_{j-1} F_j^\mathsf{T} + Q_j. \tag{19}$$

Thus $P'_{xx} = \overline{F}_M P_{xx} \overline{F}_M^\mathsf{T} + \overline{Q}_M$. The joint hook checks that this expression agrees with the covariance actually returned by the native process. With $\Phi = \mathrm{blkdiag}(\overline{F}_M, I_9)$,

$$P' = \Phi P \Phi^\mathsf{T} + \mathrm{blkdiag}(\overline{Q}_M, 0_9). \tag{20}$$

Clone and plane have no additional process noise within a pair.

The corrected research revision defines the initialized nominal state at the initialization measurement epoch under the stationary-bootstrap model. It makes no claim to have integrated the preceding initialization interval. The propagation origin must thereafter be finite and nonregressing. A camera epoch without a new LiDAR snapshot still propagates the actual intervening IMU samples; a VIO call at the same epoch adds no duplicate inertial integration. Existing map support may be used by VIO, but no old cloud is counted as a fresh LiDAR measurement. These revisions are shared by the reported matched modes, not retrospectively assigned to historical executable revisions.

C. Range/Clone Augmentation Without a Known Ground Height

At the first epoch of a pair, the native marginal is $(\mathbf{x}, P_{xx})$. The range sensor has body origin $\mathbf{t}_r$ and unit body direction $\mathbf{d}_r$. A newly observed ρ yields

$$\mathbf{l}_r = \mathbf{t}_r + \rho \mathbf{d}_r, \qquad \mathbf{q} = \mathbf{p} + R\mathbf{l}_r, \qquad h = \mathbf{n}^\mathsf{T}\mathbf{q}. \tag{21}$$

Set $(R_0, \mathbf{p}_0) = (R, \mathbf{p})$ and use the declared slope prior $(a, b) \sim \mathcal{N}((0,0), \Sigma_{ab})$. The model assumes this prior and the new range noise are independent of the native prior. It does not assume h independent of the native pose. For a right perturbation,

$$\frac{\partial h}{\partial \delta\boldsymbol{\theta}} = -\mathbf{n}^\mathsf{T} R [\mathbf{l}_r]_\times, \quad \frac{\partial h}{\partial \delta \mathbf{p}} = \mathbf{n}^\mathsf{T}, \quad \frac{\partial h}{\partial (a,b)} = (-q_x, -q_y), \quad \frac{\partial h}{\partial \rho} = \mathbf{n}^\mathsf{T} R \mathbf{d}_r. \tag{22}$$

Denote the first two nonzero derivatives embedded into a 1×19 row by J_h, and let $E_6 = [I_6 \; 0_{6\times13}]$. For $\boldsymbol{\xi} = [\delta\mathbf{x}^\mathsf{T}, \delta a, \delta b, \delta\rho]^\mathsf{T} \in \mathbb{R}^{22}$, the augmentation Jacobian is

$$B = \frac{\partial \delta \mathbf{y}}{\partial \boldsymbol{\xi}} = \begin{bmatrix} I_{19} & 0_{19\times2} & 0_{19\times1} \\ E_6 & 0_{6\times2} & 0_{6\times1} \\ 0_{2\times19} & I_2 & 0_{2\times1} \\ J_h & \begin{bmatrix} -q_x & -q_y \end{bmatrix} & \mathbf{n}^\mathsf{T} R \mathbf{d}_r \end{bmatrix}, \qquad C_\xi = \mathrm{blkdiag}(P_{xx}, \Sigma_{ab}, \sigma_\rho^2). \tag{23}$$

Then $P_0 = B C_\xi B^\mathsf{T}$. For every vector $\mathbf{v}$, $\mathbf{v}^\mathsf{T} P_0 \mathbf{v} = (B^\mathsf{T}\mathbf{v})^\mathsf{T} C_\xi (B^\mathsf{T}\mathbf{v}) \geq 0$. The native block is unchanged, while native–clone–plane cross blocks are induced by the same linear map. The cloned pose duplicates a random variable, so a singular augmented covariance is expected and must not be repaired by arbitrary diagonal noise.

The world height is therefore an uncertain nuisance variable derived from the observation and estimated pose, not a simulator constant. The seed range is not subsequently reused as a second independent observation. At a later pair, the implementation retains the native marginal and tagged gyro marginal but replaces the previous auxiliary variables with a new clone and locally initialized plane. Independent slope reinitialization is a local terrain approximation, not persistent mapping of a globally known ground surface.

D. Ray–Plane Projection and All Nonzero Jacobian Blocks

For intrinsics (f_x, f_y, c_x, c_y), define

$$\boldsymbol{\nu}(\mathbf{u}_0) = \begin{bmatrix} (u_0 - c_x)/f_x \\ (v_0 - c_y)/f_y \\ 1 \end{bmatrix}, \qquad N_u = \frac{\partial \boldsymbol{\nu}}{\partial \mathbf{u}_0} = \begin{bmatrix} 1/f_x & 0 \\ 0 & 1/f_y \\ 0 & 0 \end{bmatrix}. \tag{24}$$

The ray is not unit-normalized: λ is the first optical-axis depth. Set

$$\begin{aligned} &\mathbf{o} = \mathbf{p}_0 + R_0 \mathbf{t}_c, \qquad \mathbf{d} = R_0 R_{bc} \boldsymbol{\nu}, \quad D = \mathbf{n}^\mathsf{T}\mathbf{d}, \\ &\lambda = (h - \mathbf{n}^\mathsf{T}\mathbf{o})/D, \quad \mathbf{q} = \mathbf{o} + \lambda \mathbf{d}, \quad A_\pi = I_3 - \mathbf{d}\mathbf{n}^\mathsf{T}/D. \end{aligned} \tag{25}$$

Differentiating $\mathbf{n}^{\mathsf{T}}\mathbf{q} = h$ gives the useful identity

$$\mathrm{d}\mathbf{q} = A_\pi \mathrm{d}\mathbf{o} + \lambda A_\pi \mathrm{d}\mathbf{d} + \frac{\mathbf{d}}{D}(\mathrm{d}h - \mathrm{d}\mathbf{n}^{\mathsf{T}}\mathbf{q}). \tag{26}$$

For the current pose, introduce

$$\mathbf{k} = R_1^{\mathsf{T}}(\mathbf{q} - \mathbf{p}_1), \quad \mathbf{c} = R_{bc}^{\mathsf{T}}(\mathbf{k} - \mathbf{t}_c), \quad C = R_{bc}^{\mathsf{T}} R_1^{\mathsf{T}}, \quad \widehat{\mathbf{u}}_1 = \begin{bmatrix} f_x c_x/c_z + c_x^{\mathrm{int}} \\ f_y c_y/c_z + c_y^{\mathrm{int}} \end{bmatrix}. \tag{27}$$

Here $c_x^{\mathrm{int}}, c_y^{\mathrm{int}}$ denote principal-point coordinates, to distinguish them from components of $\mathbf{c}$. The projection derivative is

$$J = \frac{\partial \operatorname{proj}(\mathbf{c})}{\partial \mathbf{c}} = \begin{bmatrix} f_x/c_z & 0 & -f_x c_x/c_z^2 \\ 0 & f_y/c_z & -f_y c_y/c_z^2 \end{bmatrix}. \tag{28}$$

The complete nonzero blocks of the 2×28 prediction Jacobian are

$$\begin{aligned} H_{\theta_1} &= J R_{bc}^{\mathsf{T}} [\mathbf{k}]_\times, & H_{p_1} &= -JC, \\ H_{\theta_0} &= -JCA_\pi R_0 [\mathbf{t}_c + \lambda R_{bc}\boldsymbol{\nu}]_\times, & H_{p_0} &= JCA_\pi, \\ H_a &= JC\mathbf{d}\, q_x/D, & H_b &= JC\mathbf{d}\, q_y/D, \quad H_h = JC\mathbf{d}/D. \end{aligned} \tag{29}$$

Thus, in implementation order,

$$H = [H_{\theta_1}\ H_{p_1}\ 0_{2\times 13}\ H_{\theta_0}\ H_{p_0}\ H_a\ H_b\ H_h]. \tag{30}$$

There are thirteen zero native columns: inverse exposure, velocity, two biases, and gravity. The first-endpoint pixel derivative is

$$U = \frac{\partial \widehat{\mathbf{u}}_1}{\partial \mathbf{u}_0} = JC\lambda A_\pi R_0 R_{bc} N_u. \tag{31}$$

The signs follow $\mathrm{d}(R\mathbf{s}) = -R[\mathbf{s}]_\times \delta\boldsymbol{\theta}$ for a fixed body vector and $\mathrm{d}(R^{\mathsf{T}}\mathbf{s}) = [R^{\mathsf{T}}\mathbf{s}]_\times \delta\boldsymbol{\theta}$ for a fixed world vector. These identities, combined with (26), yield (29); the implemented analytic blocks are also checked against independent central differences.

For independent raw endpoint errors $\boldsymbol{\epsilon}_0, \boldsymbol{\epsilon}_1$, linearizing the residual using the observed first pixel yields

$$\mathbf{r} = H\delta\mathbf{y} + \boldsymbol{\epsilon}_1 - U\boldsymbol{\epsilon}_0, \qquad \mathcal{R} = \Sigma_1 + U\Sigma_0 U^{\mathsf{T}}. \tag{32}$$

The residual is the observed second pixel minus its prediction, not a gyro-precompensated angular flow. Correlation due to shared poses, plane, and inertial propagation resides in HPH^{T}. The measurement model assumes independent endpoint errors and a block-diagonal $\mathcal{R}$ across correspondences; correlations introduced by the image tracker itself are not identified from these data. In particular, the supplied $1\,\mathrm{pixel}$ endpoint standard deviation is an engineering input rather than a calibrated protection model.

As a geometric limiting case, suppose both optical centres coincide. Then $\mathbf{c} = \lambda\mathbf{s}$ for a fixed direction $\mathbf{s}$, and $J\mathbf{c} = 0$. Hence $\partial\widehat{\mathbf{u}}_1/\partial\lambda = 0$: pure rotation about the optical centre carries no depth sensitivity in this model. Rotation about the IMU with a nonzero camera lever arm need not have coincident optical centres. The implementation's minimum predicted body translation gate is therefore a conservative operating rule, not a general observability theorem for every rotational motion.

E. Extension of a Native LiDAR or Image Update

The native estimator constructs LiDAR point–plane and direct-image constraints using its map [1], [2]. We record the observation and information-update equations needed to connect those measurements to the joint hook; its native association, affine patch construction, and map management remain unchanged.

LiDAR point–plane observation. For a measured LiDAR point $\mathbf{s}_\ell^L$, let $\mathbf{s}_\ell = R_{iL}\mathbf{s}_\ell^L + \mathbf{t}_{iL}$ be its body coordinates. A matched map plane has normal $\mathbf{m}_\ell$, centre $\mathbf{c}_\ell$, and $h_\ell = \mathbf{m}_\ell^{\mathsf{T}}\mathbf{c}_\ell$. Its scalar residual and augmented nineteen-column Jacobian are

$$\begin{aligned} e_\ell &= \mathbf{m}_\ell^{\mathsf{T}}(R\mathbf{s}_\ell + \mathbf{p}) - h_\ell, \\ H_\ell &= \begin{bmatrix} -\mathbf{m}_\ell^{\mathsf{T}} R[\mathbf{s}_\ell]_\times & \mathbf{m}_\ell^{\mathsf{T}} & 0_{1\times 13} \end{bmatrix}. \end{aligned} \tag{33}$$

The implemented residual weighting uses the propagated pose $(\overline{R}, \overline{\mathbf{p}})$ to evaluate its uncertainty terms. With body point covariance $Q_{s,\ell} = R_{iL} Q_{L,\ell} R_{iL}^{\mathsf{T}}$ and the native joint normal/centre plane covariance $\Sigma_{mc,\ell} \in \mathbb{R}^{6\times 6}$,

$$\begin{aligned} \overline{\mathbf{q}}_\ell &= \overline{R}\mathbf{s}_\ell + \overline{\mathbf{p}}, \\ J_{mc,\ell} &= \begin{bmatrix} (\overline{\mathbf{q}}_\ell - \mathbf{c}_\ell)^{\mathsf{T}} & -\mathbf{m}_\ell^{\mathsf{T}} \end{bmatrix}, \\ \mathcal{R}_\ell &= 0.001 + J_{mc,\ell}\Sigma_{mc,\ell} J_{mc,\ell}^{\mathsf{T}} + \mathbf{m}_\ell^{\mathsf{T}} \overline{R} Q_{s,\ell} \overline{R}^{\mathsf{T}} \mathbf{m}_\ell. \end{aligned} \tag{34}$$

The first term is the retained native residual-variance floor, not the separate ToF standard deviation. This model contains local map-plane variance but does not recover the historical map–state cross covariance.

Frontal direct-image observation. Use R_{fi} for frontal-camera-from-IMU rotation and $\mathbf{t}_f$ for its optical centre in the body; these differ from the downward-camera extrinsics. For a fixed map landmark $\mathbf{q}_j^w$,

$$\mathbf{k}_j = R^\mathsf{T}(\mathbf{q}_j^w - \mathbf{p}), \qquad \mathbf{c}_{fj} = R_{fi}(\mathbf{k}_j - \mathbf{t}_f). \tag{35}$$

For a pixel within its native warped reference patch, the forward-composition residual is

$$e_{Ij} = \tau I_{\mathrm{cur}}(\mathcal{W}_j(\mathbf{x})) - \tau_{\mathrm{ref}} I_{\mathrm{ref},j}^{\mathrm{warp}}. \tag{36}$$

Here $\mathcal{W}_j$ denotes the native current-image sampling location; the reference affine patch and map landmark are held fixed at that linearization. Let ∇I_s include the implemented pyramid $1/s$ gradient scaling, and let $J_{\mathrm{proj},f}$ be the frontal pinhole projection derivative at the landmark centre. The implemented pose columns simplify to

$$\begin{aligned} [H_{I\theta}\ H_{Ip}] &= \tau \nabla I_s J_{\mathrm{proj},f} \left[R_{fi}[\mathbf{k}_j]_\times \quad - R_{fi}R^\mathsf{T} \right], \\ H_{I\tau} &= I_{\mathrm{cur}}(\mathcal{W}_j(\mathbf{x})), \\ H_{Ij} &= [H_{I\theta}\ H_{Ip}\ H_{I\tau}\ 0_{1\times 12}]. \end{aligned} \tag{37}$$

The exposure column is set to zero when exposure estimation is disabled, as in the reported profile. The simplification accounts for the camera lever arm: the native intermediate camera-centred rotation and translation derivative terms combine to $R_{fi}[\mathbf{k}_j]_\times$. Image gradients are evaluated per patch pixel, while the projective geometry is the native per-landmark linearization; the expression does not posit an independently reconstructed three-dimensional surface for every patch pixel. The profile uses forward composition, not the native optional inverse-composition path.

Native iterated information update. Stack native residuals and Jacobians into $(\mathbf{e}, H_n)$, with $\mathcal{R}_n = \mathrm{diag}(\mathcal{R}_\ell)$ for LiDAR or $\mathcal{R}_n = \sigma_I^2 I$ for frontal patch intensities. Let P_n be the native covariance held during an inner iteration, and $\mathbf{v}_i = \bar{\mathbf{x}} \boxminus \mathbf{x}_i$ the propagated-prior minus current native state. The implemented information form is

$$\begin{aligned} M_i &= (P_n^{-1} + H_n^\mathsf{T}\mathcal{R}_n^{-1}H_n)^{-1}, \\ G_i &= M_i H_n^\mathsf{T}\mathcal{R}_n^{-1}H_n, \\ \Delta\mathbf{x}_i &= M_i H_n^\mathsf{T}\mathcal{R}_n^{-1}(-\mathbf{e}) + \mathbf{v}_i - G_i\mathbf{v}_i, \\ \mathbf{x}_{i+1} &= \mathbf{x}_i \boxplus \Delta\mathbf{x}_i, \qquad B_x = (I_{19} - G_{\mathrm{final}})P_n. \end{aligned} \tag{38}$$

The final native covariance operation is performed after its native iteration/pyramid logic; it is not a repeated independent assimilation of the same patch at each iteration. In the image implementation, $\sigma_I^2 = \texttt{img_point_cov} = 100$ and the code stores

$$\widehat{M}_i = (H_n^\mathsf{T}H_n + \sigma_I^2 P_n^{-1})^{-1} = M_i/\sigma_I^2, \quad G_i = \widehat{M}_i H_n^\mathsf{T}H_n, \quad M_i H_n^\mathsf{T}\mathcal{R}_n^{-1} = \widehat{M}_i H_n^\mathsf{T}. \tag{39}$$

Thus the intensity-noise scaling is algebraically equivalent to (38), and is distinct from the new raw-flow endpoint variance of one pixel squared. The native internal $\boxminus$ retains its small-angle logarithm approximation: for $K = R - R^\mathsf{T}$, it uses $\frac{1}{2}\,\mathrm{vee}(K)$ below its small-angle threshold, and $\frac{\vartheta}{2\sin\vartheta}\,\mathrm{vee}(K)$ otherwise. The joint hook instead obtains its returned native rotation difference from the angle-axis logarithm. These first-order native iteration, fixed-map weighting, and tangent-coordinate conventions are inherited approximations, not new guarantees of nonlinear consistency.

The extension uses the native returned nominal state and covariance; it does not claim to reconstruct the native patch/map history or its unrepresented error dependencies. We next extend this returned marginal to the nine auxiliary variables.

Partition the joint prior into native $\mathbf{x}$ and auxiliary $\mathbf{z} = (\boldsymbol{\theta}_0, \mathbf{p}_0, \boldsymbol{\pi})$:

$$P^- = \begin{bmatrix} A & P_{xz} \\ P_{zx} & P_{zz} \end{bmatrix}, \qquad D = P_{zx}A^{-1}, \qquad V = P_{zz} - DAD^\mathsf{T}. \tag{40}$$

The native block A must be positive definite. Under a Gaussian linearization, $\delta\mathbf{z} = D\delta\mathbf{x} + \boldsymbol{\eta}$ with $\mathrm{Cov}(\boldsymbol{\eta}) = V$ and $\boldsymbol{\eta}$ independent of $\delta\mathbf{x}$. Suppose a native update, treated as independent of $\boldsymbol{\eta}$, returns correction $\Delta\mathbf{x}$ and pre-reset covariance $B_x \succ 0$, with $A - B_x \succeq 0$. Conditional extension gives

$$\Delta\mathbf{y} = \begin{bmatrix} \Delta\mathbf{x} \\ D\Delta\mathbf{x} \end{bmatrix}, \qquad \widetilde{P}^+ = \begin{bmatrix} B_x & B_x D^\mathsf{T} \\ DB_x & V + DB_x D^\mathsf{T} \end{bmatrix}. \tag{41}$$

This follows directly by preserving the conditional residual $\boldsymbol{\eta}$ while replacing the native marginal. In particular, the lower-right block equals $P_{zz} - D(A - B_x)D^\mathsf{T}$. The decomposition also proves positive semidefiniteness without requiring the singular full joint covariance to be inverted.

The hook treats the returned native covariance as expressed in common pre-reset local error coordinates, computes the native nominal difference in (12), injects its conditional auxiliary correction, and then applies the two-pose reset below. This is a first-order extension of the native returned marginal. It does not establish exact consistency of an iterated map-based estimator, reconstruct second-order tangent effects, or recover correlations with old map points and photometric references. Invalid, noncontractive, or mismatched returned marginals terminate the research integration rather than being silently regularized.

F. Joint Measurement Update, Optional New Range, and Reset

For a batch of N selected tracks, stack $H_i, \mathbf{r}_i$ and use $\mathcal{R} = \mathrm{blkdiag}(\mathcal{R}_1, \dots, \mathcal{R}_N)$. The one-step linearized joint EKF is

$$\begin{aligned} S &= HP^-H^\mathsf{T} + \mathcal{R}, \quad & K &= P^-H^\mathsf{T}S^{-1}, & \Delta\mathbf{y} &= K\mathbf{r}, \\ L &= I_{28} - KH, & \widetilde{P}^+ &= LP^-L^\mathsf{T} + K\mathcal{R}K^\mathsf{T}. \end{aligned} \tag{42}$$

The implementation uses positive-definite solves, not an explicit inverse of the full joint covariance. Native additive components, clone position, and plane parameters receive their corresponding increments; the native and clone rotations receive right-multiplicative increments. A frame is rejected before injection if the covariance, innovation, correction size, or resulting local plane is invalid.

For completeness, the core also implements a separate range observation for a *new* range sample. It is not used to re-fuse a pair's seed range in the reported profile. For

$$\mathbf{o}_r = \mathbf{p}_1 + R_1\mathbf{t}_r, \quad \mathbf{d}_r^w = R_1\mathbf{d}_r, \quad D_r = \mathbf{n}^\mathsf{T}\mathbf{d}_r^w, \quad \widehat{\rho} = (h - \mathbf{n}^\mathsf{T}\mathbf{o}_r)/D_r, \quad \mathbf{q}_r = \mathbf{o}_r + \widehat{\rho}\mathbf{d}_r^w, \tag{43}$$

the nonzero prediction derivatives are

$$\begin{aligned} H^\rho_{\theta_1} &= \mathbf{n}^\mathsf{T}R_1[\mathbf{t}_r + \widehat{\rho}\mathbf{d}_r]_\times/D_r, & H^\rho_{p_1} &= -\mathbf{n}^\mathsf{T}/D_r, \\ H^\rho_a &= q_{r,x}/D_r, & H^\rho_b &= q_{r,y}/D_r, \quad H^\rho_h = 1/D_r. \end{aligned} \tag{44}$$

The residual is $\rho_m - \widehat{\rho}$ with the supplied positive range variance. Unknown range or pixel covariances are rejected.

For a rotation injection $\mathbf{d}$, the right-local covariance reset is

$$J_r(\mathbf{d}) = I_3 - \frac{1-\cos\vartheta}{\vartheta^2}[\mathbf{d}]_\times + \frac{\vartheta - \sin\vartheta}{\vartheta^3}[\mathbf{d}]_\times^2, \quad \vartheta = \|\mathbf{d}\|, \tag{45}$$

with the small-angle series $I_3 - \frac{1}{2}[\mathbf{d}]_\times + \frac{1}{6}[\mathbf{d}]_\times^2$. Indeed, $\mathrm{Log}(\mathrm{Exp}(-\mathbf{d})\,\mathrm{Exp}(\mathbf{d}+\boldsymbol{\epsilon})) = J_r(\mathbf{d})\boldsymbol{\epsilon} + O(\|\boldsymbol{\epsilon}\|^2)$. Thus the reset for (13) is

$$\Gamma = \mathrm{blkdiag}\left(J_r(\Delta\boldsymbol{\theta}_1), I_{16}, J_r(\Delta\boldsymbol{\theta}_0), I_6\right), \qquad P^+ = \Gamma\widetilde{P}^+\Gamma^\mathsf{T}. \tag{46}$$

The native rotation has already been injected when a native update reaches the hook, so it is not injected twice. Its covariance reset is nevertheless applied here, together with the newly injected clone correction. The resulting 19×19 native marginal is written back to the native state. Equations (42)–(46) preserve positive semidefiniteness because sums and congruences of positive semidefinite matrices remain positive semidefinite.

G. Tagged Gyro Component and Its Exact Implemented Transforms

Write $P = P_g + P_o$ with $P_g \succeq 0$ and $P_o \succeq 0$. This is a diagnostic covariance decomposition, not an additional state or an identified physical source separation for every estimator error. At the initialized native epoch, let P_{bb} be the gyro-bias 3×3 block, and initialize

$$P_{g,x} = P_{xb_g}P_{bb}^{-1}P_{b_gx}. \tag{47}$$

This is the covariance component generated by the initial gyro-bias random variable in its Gaussian conditional decomposition. The Schur complement gives $P_{xx} - P_{g,x} \succeq 0$. It includes existing correlations with that variable, but not all initial attitude uncertainty by definition. The first ready native call is required to be the unpropagated initialization return ($F = I, Q = Q_g = 0$), so process noise preceding the tag is not silently omitted.

For each native substep, the tagged noise retains only the implemented gyro-measurement and gyro-bias process blocks:

$$Q_g = \Delta t^2\,\mathrm{blkdiag}(D_\omega, 0_3, 0, 0_3, D_{b_g}, 0_3, 0_3). \tag{48}$$

It is accumulated with the same F_j recursion as (19). The current gyro-tag native marginal is inherited when a new pair begins; it is not reinitialized to zero at every clone. With B_x^{aug} the first nineteen columns of the augmentation B,

$$P_{g,0} = B_x^{\mathrm{aug}}P_{g,x}(B_x^{\mathrm{aug}})^\mathsf{T}, \qquad P_g' = \Phi P_g\Phi^\mathsf{T} + \mathrm{blkdiag}(\overline{Q}_g, 0_9). \tag{49}$$

Slope-prior and newly acquired range noise are not tagged as gyro noise.

For a native conditional update, use D from (40) and $T = B_x A^{-1}$. The error transformation and tag are

$$L_n = \begin{bmatrix} T & 0_{19\times 9} \\ D(T - I_{19}) & I_9 \end{bmatrix}, \qquad P_g^+ = \Gamma L_n P_g^- L_n^\mathsf{T} \Gamma^\mathsf{T}. \tag{50}$$

Before any clone exists, the analogous native-only transform is $P_{g,x}^+ = J_x T P_{g,x}^- T^\mathsf{T} J_x^\mathsf{T}$, where $J_x = \mathrm{blkdiag}(J_r(\Delta\boldsymbol{\theta}_1), I_{16})$. For a joint measurement update,

$$P_g^+ = \Gamma L P_g^- L^\mathsf{T} \Gamma^\mathsf{T}, \qquad L = I - KH. \tag{51}$$

The independent new measurement-noise term $K\mathcal{R}K^\mathsf{T}$ is assigned to P_o, not to P_g. This assignment, the local noise model, and the native independence approximation define the meaning of the tag.

The decomposition remains positive semidefinite under these transforms. For (51), the remaining part is $\Gamma(LP_o^- L^\mathsf{T} + K\mathcal{R}K^\mathsf{T})\Gamma^\mathsf{T}$. For a native contraction, represent the update as a native error transform T with added covariance

$$N_x = B_x - B_x A^{-1} B_x \succeq 0. \tag{52}$$

The inequality follows from $0 \prec B_x \preceq A$ by congruence. The augmented added covariance is $[I_{19}\ D^\mathsf{T}]^\mathsf{T} N_x [I_{19}\ D^\mathsf{T}]$, again positive semidefinite and assigned to the untagged part. These are algebraic statements within the assumed local Gaussian model; they do not prove calibration against real sensor error.

H. Precisely Defined Cross-Covariance Ablations

For a partition $\mathcal{G}$, let $\mathcal{D}_\mathcal{G}(M)$ retain the principal diagonal blocks of M and zero its off-partition blocks. This positive linear map preserves positive semidefiniteness. The gyro-cross ablation, applied before the raw flow update, uses groups $(\mathbf{x}_{19}, \mathrm{clone}_6, \boldsymbol{\pi}_3)$:

$$\begin{aligned} P_g' &= \mathcal{D}_{19|6|3}(P_g), \\ P' &= P - P_g + P_g'. \end{aligned} \tag{53}$$

It removes only the tagged cross component between those three groups. Native internal gyro-bias relationships, non-gyro clone correlations, all group marginals, and the nonzero gyro tag marginals remain. The plane-cross ablation instead separates the first twenty-five variables from the plane:

$$P' = \mathcal{D}_{25|3}(P), \qquad P_g' = \mathcal{D}_{25|3}(P_g). \tag{54}$$

It removes all plane cross terms, including their tagged part, while preserving the native/clone block and the plane marginal. It is not equivalent to assuming zero range variance or a known plane height. For (53), $P' - P_g' = P - P_g \succeq 0$; for (54), $P' - P_g' = \mathcal{D}_{25|3}(P - P_g) \succeq 0$. Consequently the adjusted tag can be propagated into later pairs without discarding the initial gyro-bias component. Both variants deliberately change the statistical model. Their validity as implementations does not imply that their physical uncertainty assumptions are correct or that one variant must have the smallest trajectory error on every recording.

I. Operating Domain, Selection, and Failure Semantics

The implemented local model has the following fixed engineering domain. First and second optical depths must lie in $[0.15, 12]\,\mathrm{m}$. The normalized absolute ray–plane cosine must be at least 0.25; each ray intersection must remain within $2\,\mathrm{m}$ horizontally of the seed intersection. The slope chart must satisfy $|a|, |b| \le 1$. The plane is not silently extended over arbitrary walls, steps, curved terrain, or surfaces at different heights. Passing these geometric tests does not establish that all visible features physically lie on one plane.

The frontend limits detection to eighty features and forms raw, nonoverlapping correspondences with a 21×21 pyramidal Lucas–Kanade window and maximum pyramid level 3. Its forward/backward check is 1 pixel, with an eight-pixel image margin and the implemented OpenCV photometric-error threshold 20. That last threshold is a tracker error quantity, not a stated twenty-pixel geometric error. The joint batch requires $8 \le N \le 80$, unique track identities, valid positive endpoint covariances, and a minimum first-image span of 15 pixels on both axes after individual selection. The predicted native body displacement from the clone must be at least $0.005\,\mathrm{m}$. Defining $H_{p,i}$ as the current-position columns of H_i, the translation-information check is

$$\lambda_{\min}\left(\sum_{i=1}^{N} H_{p,i}^\mathsf{T} \mathcal{R}_i^{-1} H_{p,i}\right) \ge 10^{-8}. \tag{55}$$

This check and the displacement rule are practical degeneracy filters; they do not replace a full nonlinear observability analysis.

Each candidate track uses the prior innovation covariance $S_i = H_i P H_i^\mathsf{T} + \mathcal{R}_i$ and is retained only if

$$\mathbf{r}_i^\mathsf{T} S_i^{-1} \mathbf{r}_i \le 9.210340371976184. \tag{56}$$

The value is the nominal 99% two-dimensional chi-square cutoff, but the supplied pixel covariance and feature-selection process are not empirically calibrated to imply that coverage. For the final batch of $m = 2N$ scalar residuals, require

$$\mathbf{r}^{\mathsf{T}} S^{-1} \mathbf{r} \le 3m. \tag{57}$$

The factor $3m$ is explicitly an engineering bound, not an exact chi-square quantile. The optional new-range update uses the nominal one-dimensional cutoff 6.6348966010212145. For a joint flow or new-range measurement, before injection both rotation corrections must be at most 0.2 radians and both position corrections at most $0.5\,\mathrm{m}$. The corrected plane must remain in its slope domain. Invalid covariances, nonfinite values, and failed positive-definite innovation solves are rejected without applying the joint measurement correction. Symmetry and positive semidefiniteness of P, P_g, and $P - P_g$ are checked with finite-precision tolerances; missing or invalid covariance inputs at the core API are rejected. The caller explicitly supplies the declared, uncalibrated pixel model.

The reported acquisition-preserving integration requires an exact common front/down-camera/range seed timestamp; it does not choose a nearest range and relabel its acquisition time. A seed is paired with the next camera epoch, with a maximum interval of 0.3 seconds. A missing image, invalid seed, excessive interval, insufficient features, or failed gate is logged with its original acquisition identity. All five input counts and stream tails, including the final publication barrier, must be accounted for before processing completion is reported. A final seed lacking a second image is a pending, right-censored pair; complete input accounting does not mean that every pair produced an accepted state update.

J. Scope of a Retained Serial Prototype

A separate engineering comparison has been prototyped but is *not integrated into the native replay cohort reported in this draft*. It is therefore a planned comparison, not an experimental result here. At fixed, uncertain native attitude, cloned pose, and plane means, it fits the current world position to the same raw pixels using a bounded robust reprojection optimization. With position Jacobian J_p, nuisance Jacobian H_n, and all mutual cross terms retained in the twelve-dimensional nuisance marginal P_{nn}, its first-order measurement covariance is

$$A_s = (J_p^{\mathsf{T}} W J_p)^{-1} J_p^{\mathsf{T}} W, \qquad B_s = -A_s H_n, \qquad R_s = A_s \mathcal{R} A_s^{\mathsf{T}} + B_s P_{nn} B_s^{\mathsf{T}}. \tag{58}$$

The resulting position is then proposed as a native nineteen-dimensional position-measurement update with the same auxiliary conditioning and rotation-reset convention. The serial boundary deliberately assumes that this derived position measurement is independent of the native state, although shared estimation variables generally violate that assumption. Neither its attitude nor its range-derived height is treated as ground truth. Its robust-selection covariance remains a first-order model, and standalone checks do not establish integration, superiority, or hardware performance. No result in the present experimental cohort is attributed to this unintegrated prototype.

APPENDIX B
IMPLEMENTED IMPACT PLANNING AND EXECUTION MATHEMATICS

This appendix documents the existing task layer rather than an additional experiment with the new raw-flow estimator. The directional-information, empirical-calibration, candidate-selection and fault-response implementations were developed and evaluated in the historical FAST-LIO/P6–P15 simulation chain. Their equations are included to specify the intended estimator-to-task interface. They have not been jointly revalidated with the new FAST-LIVO2 clone–plane–flow state. In particular, positive-semidefinite Gaussian covariance from the new filter is not a calibrated protection level (PL), and the old calibration must not be transferred by substituting that covariance. In this appendix $[x]_+ = \max(x, 0)$, $\mathrm{clip}(x, a, b)$ clips a scalar, and all obstacle and surfel vectors use the declared map frame.

A. Observed Directional Information and Inflation

For accepted point-to-plane constraints with normals $\mathbf{n}_i$, residuals r_i, geometry qualities g_i, and configured residual scale $s_r > 0$, `information_from_constraints` in `integrity.py` implements

$$\bar{\mathbf{n}}_i = \mathbf{n}_i / \|\mathbf{n}_i\|, \qquad w_i = \frac{\mathrm{clip}(g_i, 0, 1)}{1 + (|r_i|/s_r)^2}, \tag{59}$$

$$\Lambda_t = \sum_i w_i \bar{\mathbf{n}}_i \bar{\mathbf{n}}_i^{\top}. \tag{60}$$

Zero-length normals contribute zero. Matrices are symmetrized numerically. The optional temporal memory is disabled when its horizon $h = 0$; otherwise, for advancing stamps,

$$\rho_t = \exp[-(t - t_-)/h], \qquad \widetilde{\Lambda}_t = \rho_t \widetilde{\Lambda}_{t-} + \Lambda_t, \qquad \widetilde{N}_t = \rho_t \widetilde{N}_{t-} + 1. \tag{61}$$

If $\widetilde{N}_t > N_{\max}$, both accumulated information and the equivalent frame count are multiplied by $N_{\max}/\widetilde{N}_t$. Initialization, nonadvancing stamps, or $h = 0$ reset memory to the current matrix and one frame. This bounded geometric memory is not a proof that revisited constraints are statistically independent.

Let $\lambda_1 \le \lambda_2 \le \lambda_3$ be the nonnegative-clipped eigenvalues used by the diagnostic and $\mathbf{q}_1$ its weakest eigenvector. For configured $\epsilon > 0$, reference condition number $c_0 > 1$, and cap $D_{\max}$, the implemented inflation is

$$c_t = \frac{\lambda_3}{\max(\lambda_1, \epsilon)}, \qquad D_t = \operatorname{clip}\left(\frac{\log \max(c_t, 1)}{\log c_0}, 0, D_{\max}\right), \tag{62}$$

$$\kappa_t = 1 + a_D D_t + a_N[\xi_N]_+ + a_T[\xi_T]_+ + a_R[\xi_R]_+, \tag{63}$$

$$P_t^{\text{int}} = \kappa_t \mathcal{E}_{p_{\min}}(P_{p,t}^{\text{LIO}}) + \eta(\widetilde{\Lambda}_t + \epsilon I)^{-1}, \tag{64}$$

$$\mathrm{PL}_t(\mathbf{d}) = K\sqrt{[\mathbf{d}^\top P_t^{\text{int}} \mathbf{d}]_+}, \qquad \|\mathbf{d}\| = 1. \tag{65}$$

Here $\mathcal{E}_{p_{\min}}$ symmetrizes a covariance and floors its eigenvalues at $p_{\min} > 0$. Axis diagnostics use $\mathbf{d} = \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$; the weak-axis diagnostic uses $\mathbf{q}_1$. The current P6 wrapper exposes ξ_N, ξ_T, ξ_R as configured diagnostic inputs, with defaults zero; it does not estimate all three online from innovations or learn the inflation coefficients. The regularizer and inflation are engineering choices whose calibration is estimator dependent.

B. Frozen Empirical Calibration and Its Domain

The P7 collector aligns estimate and truth once using the first matched position and yaw. With this fixed rotation R_0, it evaluates

$$\tilde{\mathbf{p}}_i = R_0(\mathbf{p}_i - \mathbf{p}_0) + \mathbf{p}_0^{\text{ref}}, \qquad \mathbf{e}_i = \mathbf{p}_i^{\text{ref}} - \tilde{\mathbf{p}}_i, \tag{66}$$

$$P_i' = R_0 P_i^{\text{int}} R_0^\top, \qquad z_{i,d} = \frac{|\mathbf{d}_i^\top \mathbf{e}_i|}{\sqrt{\max(\mathbf{d}_i^\top P_i' \mathbf{d}_i, 10^{-18})}}. \tag{67}$$

The truth trajectory is interpolated only in the evaluator; it is not an input to the predictor. The four calibrated directions are the three map axes and the rotated instantaneous weak direction. For each training scenario s and direction d, `calibrate_p7.py` computes the *higher* empirical quantile

$$q_{\alpha,s,d} = z_{(1+\lceil (n_{s,d}-1)\alpha \rceil),s,d}, \qquad \alpha \in \{0.95, 0.99\}, \tag{68}$$

where parentheses denote ascending order statistics. The train-only common shift reserve and directional factors are

$$\chi = \max\left(1, \max_d \frac{\max_s q_{0.95,s,d}}{\max(\min_s q_{0.95,s,d}, 10^{-12})}\right), \tag{69}$$

$$K_{\alpha,d} = \chi \max_s q_{\alpha,s,d}, \qquad K_{\text{plan}} = \max_d K_{0.95,d}. \tag{70}$$

P9 and P11 verify a frozen train-only artifact and use K_{plan} for obstacle directions. Held-out coverage is the observed fraction $n^{-1}\sum_i \mathbf{1}[z_{i,d} \le K_{\alpha,d}]$. Neither the cross-scenario multiplier nor taking the maximum over four directions proves distribution-free coverage for arbitrary directions, simultaneous trajectories, future environments, or temporally dependent samples. These are conditional empirical results on the old estimator and data split, not a certified failure-probability bound. Recalibration and new held-out evaluation are required for the joint raw-flow estimator.

C. Sampled Geometry, Delay Reserves and Braking Helpers

`alert_limit.py` samples a B-spline using de Boor recursion. For degree p, knots u_i, and span $u \in [u_i, u_{i+1})$, the recursion is

$$\mathbf{b}_j^{(0)} = \mathbf{c}_j, \qquad \mathbf{b}_j^{(r)} = (1 - \alpha_{j,r})\mathbf{b}_{j-1}^{(r-1)} + \alpha_{j,r}\mathbf{b}_j^{(r-1)}, \tag{71}$$

$$\alpha_{j,r} = \frac{u - u_j}{u_{j+p-r+1} - u_j}, \qquad r = 1, \dots, p, \quad j = i - p + r, \dots, i. \tag{72}$$

The initial control indices are $j = i - p, \dots, i$, and the evaluated point is $\mathbf{x}(u) = \mathbf{b}_i^{(p)}$. The implementation uses zero for $\alpha_{j,r}$ when its denominator is at most 10^{-15}, samples the valid remaining domain at the configured interval, and appends its final endpoint. For sampled positions $\mathbf{x}_k$ and observed obstacle points $\mathcal{O}$, define

$$\mathbf{o}_k = \arg\min_{\mathbf{o} \in \mathcal{O}} \|\mathbf{o} - \mathbf{x}_k\|, \qquad d_k = \|\mathbf{o}_k - \mathbf{x}_k\|, \quad \mathbf{a}_k = (\mathbf{o}_k - \mathbf{x}_k)/d_k, \tag{73}$$

$$r_{\text{delay}}(v, L) = vL + \tfrac{1}{2} a_{\max} L^2, \tag{74}$$

$$\mathrm{AL}_{c,k} = d_k - r_{\text{body}} - r_{\text{base}} - r_{\text{track}} - r_{\text{dynamic}} - r_{\text{delay}}(v_c, L). \tag{75}$$

Zero-clearance samples cannot supply a valid unit obstacle direction. Empty, nonfinite or misaligned inputs are not interpreted as free space. The P11 wrapper obtains v_c from sampled path length divided by duration; its dynamic reserve defaults to zero. This is a sampled point-map test, not continuous swept-volume collision checking.

The separate legacy grid-proxy helpers in `planning.py` implement

$$r_{\mathrm{grid}} = r_{\mathrm{body}} + r_{\mathrm{base}} + k_\sigma \sqrt{[\lambda_{\max}(P_{xy})]_+} + [v]_+[L]_+ + [r_{\mathrm{dynamic}}]_+, \tag{76}$$

$$d_{\mathrm{brake}} = \frac{[v]_+^2}{2a_{\mathrm{dec}}} + [v]_+[L]_+ + r_{\mathrm{brake}}, \qquad a_{\mathrm{dec}} > 0. \tag{77}$$

Grid inflation uses $\lceil r_{\mathrm{grid}}/\Delta_{\mathrm{grid}} \rceil$ cells and rejects diagonal moves that cut occupied corners. These helper formulas must not be silently added to Eq. (75): the latter does not contain an additional $v^2/(2a_{\mathrm{dec}})$ term. The actual controller braking response and attainable deceleration require separate validation.

D. Hard Directional Margin and Perception Forecast

For each remaining trajectory sample the implemented hard test is

$$\mathrm{PL}_{c,k} = K_{\mathrm{plan}} \sqrt{[\mathbf{a}_{c,k}^\top \hat{P}_{c,k} \mathbf{a}_{c,k}]_+}, \qquad M_{c,k} = \mathrm{AL}_{c,k} - \mathrm{PL}_{c,k}, \tag{78}$$

$$M_c = \min_k M_{c,k}, \qquad F_c^{\mathrm{int}} = \mathbf{1}[M_c \geq m_{\mathrm{reserve}}]. \tag{79}$$

P9 uses the current P^{int} at every sample. P10/P11 instead use a candidate-specific forecast from `minimum_excitation.py`. For map surfels with position $\mathbf{s}_j$, unit normal $\mathbf{n}_j$, static confidence c_j, geometry quality g_j, and age $A_j = [t - t_j^{\mathrm{seen}}]_+$, that forecast is

$$\hat{\Lambda}_{c,k} = s_I \sum_j \mathbf{1}[\|\mathbf{s}_j - \mathbf{x}_{c,k}\| \leq r_{\mathrm{vis}}] c_j g_j \exp(-A_j/\tau_I) \mathbf{n}_j \mathbf{n}_j^\top, \tag{80}$$

$$\hat{P}_{c,k} = \mathcal{E}_{p_{\min}^{\mathrm{pred}}} \left[\left((P_0^{\mathrm{int}})^{-1} + \sum_{\ell=0}^{k} \hat{\Lambda}_{c,\ell} \right)^{-1} \right]. \tag{81}$$

The inverse-variance scale s_I is configured. The default prediction floor is $p_{\min}^{\mathrm{pred}} = 10^{-12}$, distinct from the default 10^{-9} floor on the incoming LIO covariance. Visibility here is a radius test, without ray-traced occlusion. The forecast adds no process noise and can repeatedly count surfels along a path; it does not carry joint state–map or repeated-observation correlations. It is consequently a heuristic perception forecast, not the covariance propagation of the new joint filter or a calibrated future PL.

E. Task Gain, Collision Proxy and Energy Accounting

For a candidate path let $\ell_c = \sum_k \|\Delta \mathbf{x}_{c,k}\|$ and $D_c = \|\mathbf{x}_{c,\mathrm{end}} - \mathbf{x}_{c,0}\|$. The online map-derived metric in `candidate_metrics.py` is

$$b_j = g_j\{0.10 + 0.45(1 - c_j) + 0.45[1 - \exp(-A_j/\tau_G)]\}, \tag{82}$$

$$V_c = \sum_j b_j \mathbf{1}[\exists k : \|\mathbf{s}_j - \mathbf{x}_{c,k}\| \leq r_{\mathrm{vis}}], \tag{83}$$

$$\bar{V}_c = \begin{cases} (V_c - V_{\min})/(V_{\max} - V_{\min}), & V_{\max} - V_{\min} > 10^{-12}, \\ 1/2, & \text{otherwise}, \end{cases} \tag{84}$$

$$G_c = \mathrm{clip}\big(w_p \,\mathrm{clip}(D_c/\max(\ell_c, 10^{-12}), 0, 1) + (1 - w_p)\bar{V}_c, 0, 1\big). \tag{85}$$

Normalization uses only the immutable candidate batch. This is a proxy for progress efficiency and revisiting map support, not mutual information of the estimator; no candidate name determines its value in online-map mode. The controlled metadata mode remains available for mechanism ablations.

The collision metric is the maximum *pointwise* one-sided Gaussian tracking proxy

$$\hat{p}_c = \max_k \frac{1}{2} \operatorname{erfc}\left(\frac{\mathrm{AL}_{c,k} + r_{\mathrm{track}}}{\sqrt{2}\,\sigma_{\mathrm{track}}} \right), \qquad \sigma_{\mathrm{track}} = r_{\mathrm{track}}/k_{\mathrm{track}}. \tag{86}$$

Adding back the tracking reserve avoids subtracting that reserve twice. This maximum is neither a path-union collision probability nor a certified risk bound; localization uncertainty remains in the separate margin test.

For segment interval $\Delta t_c = T_c/(N_c-1)$ and $\mathbf{v}_{c,k} = \Delta\mathbf{x}_{c,k}/\Delta t_c$, the implemented energy proxies are

$$E_c^{\mathrm{inc}} = \ell_c + w_D \sum_k \|\mathbf{v}_{c,k}\|^2 \Delta t_c + w_C \sum_k [\Delta z_{c,k}]_+ + w_A \sum_k \left\| \frac{\Delta \mathbf{v}_{c,k}}{\Delta t_c} \right\|^2 \Delta t_c, \tag{87}$$

$$E_c^{\mathrm{ret}} = \|\mathbf{x}_{\mathrm{home}} - \mathbf{x}_{c,\mathrm{end}}\| + 1.5[z_{\mathrm{home}} - z_{c,\mathrm{end}}]_+. \tag{88}$$

The defaults are $w_D = 0.15$, $w_C = 1.5$, and $w_A = 0.05$. These quantities are weighted planning proxies, not measurements in joules or a guarantee that the straight-line return is traversable. The controller includes already-accounted cost E_{spent} in candidate totals. An interrupted segment with advertised total $E_{\mathrm{decision}} \geq E_{\mathrm{spent}}$ updates

$$E'_{\mathrm{spent}} = E_{\mathrm{spent}} + \mathrm{clip}(T_{\mathrm{executed}}/T_c, 0, 1)(E_{\mathrm{decision}} - E_{\mathrm{spent}}). \tag{89}$$

F. Feasibility Before Utility and Minimum Intervention

`integrity_exploration.py` forms

$$\mathcal{F} = \{c : M_c \geq m_{\mathrm{reserve}},\ \hat{p}_c \leq p_{\max},\ E_{\mathrm{spent}} + E_c^{\mathrm{inc}} + E_c^{\mathrm{ret}} \leq E_{\mathrm{budget}}\}, \tag{90}$$

$$U_c = w_G G_c - w_T T_c - w_E(E_{\mathrm{spent}} + E_c^{\mathrm{inc}}), \qquad c^* \in \arg\max_{c\in\mathcal{F}} U_c. \tag{91}$$

An empty $\mathcal{F}$ returns no selected trajectory. Integrity margin is absent from U_c: a larger predicted margin cannot compensate for a failed hard constraint. The information-only comparator still applies collision and return-energy tests but omits the integrity test. An optional declared utility-resolution band $\delta_U > 0$ first forms $\mathcal{F}_\delta = \{c \in \mathcal{F} : U_c \geq U_{\max} - \delta_U - 10^{-12}\}$. When it contains several candidates, selection lexicographically minimizes $(E_c^{\mathrm{inc}} + E_c^{\mathrm{ret}}, T_c, -U_c, \mathrm{name}_c)$; otherwise exact utility maximization is used, with deterministic name-based ties.

The separate finite-action P10 kernel generates baseline, sinusoidal lateral and vertical offsets, longer-duration variants, and backtracking. A lateral example is $\mathbf{x}_c(\tau) = \bar{\mathbf{x}}(\tau) \pm o_y \sin(\pi\tau)\mathbf{n}_\perp(\tau)$, $0 \leq \tau \leq 1$. Its velocity-squared energy surrogate and recovery objective are

$$E_c^{(10)} = \sum_k \|\Delta\mathbf{x}_{c,k}\|^2/\Delta t_c, \tag{92}$$

$$C_c = [T_c - T_0]_+ + \lambda_E \Delta E_c + \lambda_\ell \Delta \ell_c, \qquad c_{\mathrm{rec}} = \arg\min_{c:M_c \geq m_{\mathrm{reserve}}} C_c. \tag{93}$$

The baseline is retained if feasible. Otherwise the least-cost feasible member is selected; no feasible member yields no recovery. Extra distance and energy are nonnegative relative to baseline; the longer-duration `short_hover` candidate adds a configured hover-energy bias. The kernel is a discrete surrogate search, not a continuous optimal-control solution or a proof of minimum physical sensing energy.

G. Measured Latency and Runtime Intervention

P13 records acquisition age in the source ROS clock and processing intervals in a monotonic clock. Its implemented diagnostic duration is

$$L_i = [t_{i,\mathrm{recv}}^{\mathrm{ROS}} - s_i^{\mathrm{ROS}}]_+ + [m_{i,\mathrm{command}} - m_{i,\mathrm{recv}}]_+. \tag{94}$$

This mixed-domain accounting requires the declared simulation-time contract; it is not a physical sensor-to-actuator latency calibration. For sorted finite nonnegative durations, `latency_safety.py` uses the nearest-rank statistic

$$L_q = L_{(\max(1,\lceil qn \rceil))}, \qquad q \in \{0.5, 0.95, 0.99, 1\}. \tag{95}$$

The controller's default window holds 500 samples. An empty statistic is infinite; before measurements the controller explicitly uses configured planner delay plus a configured overhead as a fallback, not a measured p99. This quantile convention differs from the P7 higher quantile in Eq. (68). Solving the delay-aware margin inequality gives the implemented speed cap

$$v_{\mathrm{cap}} = \begin{cases} v_{\max}, & L \leq 0, \\ \min\left(v_{\max}, \max\left(0, \frac{d - r_{\mathrm{fixed}} - \mathrm{PL} - m_{\mathrm{reserve}} - \frac{1}{2}a_{\max}L^2}{L}\right)\right), & L > 0. \end{cases} \tag{96}$$

The generic envelope uses configured geometry, fixed buffer and PL; the optional P15 runtime monitor instead recomputes nearest-surface geometry and $\mathbf{a}^\top P^{\mathrm{int}}\mathbf{a}$ from the current map and state. If its current margin stays below the configured reserve for the confirmation dwell and the replan cooldown has elapsed, it publishes zero velocity, closes the interrupted planning window and requests a fresh immutable candidate batch. A zero command is a requested intervention, not proof of instantaneous braking or a safe hover.

When no trajectory passes the hard test and the configured local dynamic lookahead is clear, a separate bounded recovery command may use the transverse direction toward the best rejected candidate's midpoint. If unavailable, it uses the weak

eigenvector, with sign chosen away from the closest mapped surface. For fixed unit direction $\mathbf{d}_{\mathrm{rec}}$ and recovery anchor $\mathbf{p}_a$, the implemented command is

$$\mathbf{v}_{\mathrm{rec}} = \begin{cases} v_{\mathrm{rec}}^{\max}\mathbf{d}_{\mathrm{rec}}, & \mathbf{d}_{\mathrm{rec}}^{\top}(\mathbf{p}-\mathbf{p}_a) < o_{\max}, \\ \mathbf{0}, & \text{otherwise.} \end{cases} \tag{97}$$

It does not execute the rejected trajectory. The command becomes zero when its measured projected displacement reaches the configured offset; retry count is capped separately. This is not a proven physical stopping-distance bound, and the recovery command is not certified by the rejected path's margin. The historical P14 wrapper also contains a lateral oscillation comparator: its nominal sign alternates every half-period and reverses at the configured lateral bounds.

H. Priority-Ordered Fault-State Policy

The P14 supervisor receives explicit experimental fault events, not a learned hardware fault diagnosis. For the maximum active elapsed LiDAR dropout age a_L, it selects

$$S_L(a_L) = \begin{cases} \textsc{Cautious}, & 0 \le a_L < 0.5\,\mathrm{s}, \\ \textsc{Recovery}, & 0.5 \le a_L < 1.2\,\mathrm{s}, \\ \textsc{Brake}, & 1.2 \le a_L < 1.8\,\mathrm{s}, \\ \textsc{Hover}, & 1.8 \le a_L < 3.0\,\mathrm{s}, \\ \textsc{Land}, & a_L \ge 3.0\,\mathrm{s}. \end{cases} \tag{98}$$

Evaluation is ordered: manual override; active LiDAR dropout; low battery (LAND for supplied severity ≤ 0.15, otherwise RETURN); planner delay (BRAKE); odometry delay or covariance inflation (RECOVERY); IMU dropout, timestamp jitter, or CPU-load shedding (CAUTIOUS). Camera failure sets a geometry-only flag while allowing the local LiDAR/IMU loop; ground-link packet loss alone does not stop that loop. BRAKE/HOVER/MANUAL publish zero velocity in the simulation controller. The P14 LAND branch integrates a bounded descent command; it is not evidence of a physical touchdown detector. These policy thresholds describe historical simulation behavior and have not been validated together with the new flow estimator or the new AP mission.

I. AP Clock, Source Freshness and Current Receipt Gates

The AP471 interface preserves acquisition stamps s_k and separately records host wall observations w_k bracketed by monotonic samples m_k^-, m_k^+. With $m_k = (m_k^- + m_k^+)/2$, the opt-in transport mapping is

$$t_k^{\mathrm{tr}} = w_{k_0} + m_k - m_{k_0}, \qquad 0 \le m_k^+ - m_k^- \le 5\,\mathrm{ms}. \tag{99}$$

The first trusted bracket fixes one anchor. Wider uncertain brackets are rejected; invalid or nonadvancing monotonic observations latch a fault. Mapped nanoseconds and their floored microsecond identifiers must advance strictly and fit ROS time. Raw wall steps and anchor drift remain in the ledger. A zero mapped-minus-monotonic residual follows by construction and does not demonstrate UTC, acquisition-time or FCU clock calibration. The default raw-clock mode instead uses $t_k^{\mathrm{tr}} = w_k$. Its guard, also applied to the selected transport clock, requires strict positive source, transport and observation ordering and

$$|(t_k^{\mathrm{tr}} - t_{k-1}^{\mathrm{tr}}) - (m_k - m_{k-1})| \le 50\,\mathrm{ms}, \tag{100}$$

$$|(t_k^{\mathrm{tr}} - t_{k_0}^{\mathrm{tr}}) - (m_k - m_{k_0})| \le 50\,\mathrm{ms}. \tag{101}$$

These are declared engineering consistency budgets. Faults latch until a new instance is constructed; source stamps are not clamped or replayed.

Independent acquisition-clock checks require the latest advancing simulation-clock sample c to be recently received and

$$-20\,\mathrm{ms} \le c - s_k \le 350\,\mathrm{ms}, \qquad 0 \le m_{\mathrm{now}} - m_{c,\mathrm{recv}} \le 350\,\mathrm{ms}. \tag{102}$$

Repeated clock callbacks do not refresh the last advancing-clock receipt. A clock regression latches failure. Independently, accepted source receipt age must be at most 350 ms. Clock readiness requires five seconds without an excessive gap and a rate of at least 4 Hz. Over the trailing one-second half-open window with $N \ge 2$ accepted samples, that rate is

$$f_{\mathrm{ready}} = \min\left(\frac{N}{1\,\mathrm{s}}, \frac{N-1}{m_{\mathrm{last}} - m_{\mathrm{first}}}\right). \tag{103}$$

A source timeout resets the warm-up; valid warming frames may be forwarded without asserting readiness.

The startup gate requires a unique MAVROS odometry subscriber with reliable delivery and volatile durability. Its configured identity is node `odometry`, namespace /uav1/mavros, and message type nav_msgs/msg/Odometry. Sampled visibility must persist for at least one second, with observation gaps at most 250 ms, query duration at most 50 ms, and current evidence age

at most 250 ms. Only one graph worker may be outstanding. The timeout classifies and rejects late results; it is not a proven hard real-time or GIL-scheduling bound. Identity discovery proves neither internal TF readiness nor FCU receipt. Pre-gate source frames are skipped rather than queued for later transmission.

For the FCU, SYS_STATUS must come from system/component $(1, 1)$ with valid framing. The vision-position bit $b = 128$ must occur in all three base masks:

$$H_{\mathrm{FCU}} = \mathbf{1}[(B_{\mathrm{present}} \,\&\, b) = b]\, \mathbf{1}[(B_{\mathrm{enabled}} \,\&\, b) = b]\, \mathbf{1}[(B_{\mathrm{health}} \,\&\, b) = b]. \tag{104}$$

The version-specific parameter proof includes `VISO_TYPE=1`, `PLND_ENABLED=0`, and `ARMING_SKIPCHK=0`, together with the declared GNSS-off and EKF source settings. Disabling precision landing disambiguates its shared status bit. AP backend health denotes receipt within its 300 ms timeout; it is neither a per-frame acknowledgement, measurement-quality certificate, nor an EKF fusion acknowledgement [11].

Let F_{local} collect current local source/clock flags, the enabled acquisition and exact-startup checks, and absence of a latched clock fault. The implemented current receipt gate is the conjunction

$$\begin{aligned} R_{\mathrm{base}} = F_{\mathrm{local}} \;\wedge\; \Pi_{17} \;\wedge\; C_{\mathrm{FCU}} \;\wedge\; A_{\mathrm{state}} \le 2\,\mathrm{s} \;\wedge\; A_{\mathrm{local}} \le 350\,\mathrm{ms} \\ \wedge\; A_{\mathrm{advancing\ source}} \le 350\,\mathrm{ms} \;\wedge\; A_{\mathrm{SYS_STATUS}} \le 350\,\mathrm{ms} \;\wedge\; H_{\mathrm{FCU}} \;\wedge\; D_H \ge 5\,\mathrm{s}, \end{aligned} \tag{105}$$

$$R_{\mathrm{mission}} = R_{\mathrm{base}} \;\wedge\; A_{\mathrm{last\ valid\ FCU\ odom}} \le 350\,\mathrm{ms}. \tag{106}$$

All ages are nonnegative monotonic receive ages. Here Π_{17} is exact verification of the 17 required parameters; D_H is the interval spanned by actual healthy FCU samples with gaps at most 350 ms, not time accumulated by the observer timer. Accepted FCU odometry must have an advancing original header, finite position, and quaternion norm within $[0.999, 1.001]$. The mission rechecks R_{mission} immediately before each ARM/TAKEOFF dispatch and throughout its execution phases. Loss disables research execution and requests LAND cleanup. An outstanding or recently acknowledged actuation future is retained: cancelling a ROS future cannot retract a transmitted MAVLink command. Disarmed cleanup additionally requires completion of that future, a two-second observation interval, and a fresh connected post-completion state. The bounded cleanup deadline can still produce FAIL with unresolved or armed state explicitly recorded; landing success is not assumed from a command request.

J. Velocity Boundary and Verified AP471 Scope

The separate AP velocity executor accepts increasing sequence identifiers, nondecreasing generation/source stamps, configured future-skew limits and both ROS-age and monotonic-watchdog limits. For a valid enabled command, connected armed GUIDED state and healthy fresh localization, define $\mathcal{B}_v(\mathbf{u}) = \mathbf{u}\min(1, v/\|\mathbf{u}\|)$, with $\mathcal{B}_v(\mathbf{0}) = \mathbf{0}$. Its normal output is

$$\mathbf{v}_k = \mathbf{v}_{k-1} + \mathcal{B}_{a_{\max}\Delta m}\left[R_z(\psi_{\mathrm{map}\to\mathrm{FCU}})\mathcal{B}_{v_{\max}}(\mathbf{v}_k^{\mathrm{cmd}}) - \mathbf{v}_{k-1}\right]. \tag{107}$$

Command timeout with otherwise healthy localization requests immediate zero and overrides the normal slew bound; failed localization eligibility suppresses output. The first eligible step and any reset use a zero ramp origin and $\Delta m = 0$. Enable transitions require a subsequent accepted command. Clock rollback latches failure, and sequence high-water marks survive disable/enable cycles. These are software boundary conditions, not a safe-hover guarantee.

The separate AP471 preparation compiled official commit `dbe79216`. The successful disarmed smoke verified all 1,347 advertised parameters, including the 17 required settings, with 27 disarmed heartbeats. The latest isolated autonomy build passed 769 package tests; a separate ROS-only fixture passed eight graph/source/actuation-block checks. Two real, unarmed MAVROS/Gazebo/AP471 trials matched respectively 557 synthetic-source and 585 FAST-LIO-source adapter outputs to the complete ordered wire ODOMETRY and FCU VISP/VISV logs. The synthetic source was zero pose/velocity with advancing simulation acquisition stamps; it was not estimator-accuracy evidence. Its two-second source pause revoked current receipt readiness and required a new warm-up before recovery. These runs used no new raw-flow estimator in an armed control loop. Gazebo shutdown escalation remains a separate recorded issue. The lean trials neither reproduce nor establish a fix for the earlier full-load seven-frame startup gap or 4.80 s outage.

The planned physical platform uses a Livox Mid-360, camera, MicoAir downward flow module, NVIDIA Orin NX and CUAV X25 MEGA running AP 4.7.1. The camera and flow-module models, acquisition-time calibration, board timing and hardware closed-loop behavior remain future work. Historical ArduCopter 4.5.7 flight recordings, offline new-estimator updates, and these AP471 unarmed interface tests are distinct evidence and are not pooled as one completed flight system.

APPENDIX C
REPRODUCIBILITY AND EVIDENCE BOUNDARIES

The editable source package includes generated numerical tables, plotting code, figure data, evaluator summaries and an evidence index with full source locations and SHA-256 digests. These are local research artifacts, not a public benchmark

release. Raw recorded databases remain immutable in the research workspace; indexing and replay use read-only SQLite with the immutable option, and verify the complete file family before and after use. Original sensor payload hashes and acquisition headers are preserved.

The current comparison uses one input sequence with the following roles:

1) Native: no new auxiliary-state filtering or optical-flow update; common propagation-bootstrap and asynchronous-snapshot fixes are used.
2) Joint/no-flow: clone/plane bookkeeping and the same manifold reset handling as the full model, with the raw-flow correction disabled.
3) Full: the retained joint covariance and raw two-view update.
4) Zero gyro cross: only the tagged gyro covariance component is block-diagonalized across native, clone and plane partitions.
5) Zero plane cross: plane cross blocks are removed from both total and tagged gyro covariance, while marginal blocks are retained.

The primary comparison for isolating flow is Full versus Joint/no-flow. Native provides context but differs in covariance reset handling. The two cross-covariance controls are intentionally misspecified models. A numerically positive-semidefinite ablation is not automatically a physically correct stochastic model.

A. Independent Accepted-Update Reconstruction

For every accepted raw-flow batch, the evaluator links raw endpoint pixels and track identifiers to the exact source acquisition pair. It checks the state prior against the logged pre-update mean and covariance, reconstructs the ray–plane projection, and central-differences every one of the 28 error-state columns. It also checks the two-endpoint pixel covariance, $S = HPH^\top + R$, $K = PH^\top S^{-1}$, $\delta x = Kr$, the reported NIS, both rotation-reset blocks and the full Joseph posterior. The recorded native state and covariance are checked against the resulting injection. Different native-state addresses, reuse of an image endpoint, wrong correlation masks, or absent five-stream completion records fail the audit.

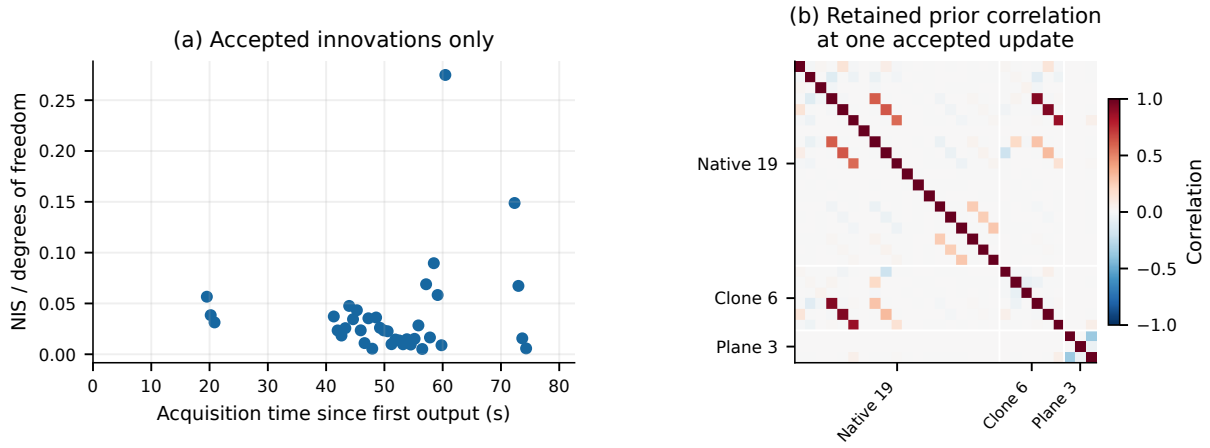


Fig. 3. Additional diagnostics from the full-model replay. Left: innovation statistics of accepted batches under the declared engineering pixel noise. Right: the normalized retained joint covariance before one accepted update. Neither plot is a test of global covariance calibration or a protection-level certificate.

Fig. 3 exposes both the selected innovation statistics and the retained off-diagonal covariance structure for inspection.

B. Interpretation of the Accuracy Metric

The separate evaluator matches original truth observations within 15 ms of a published estimate. A single initial SE(3) transform fixes the gauge; no scale fitting, per-segment alignment or later refitting is performed. For matched positions, the reported error is

$$e_k = \left\| \hat{p}_k - (R_A p_k^{\mathrm{gt}} + t_A) \right\|_2, \quad \mathrm{ATE}_{\mathrm{RMS}} = \sqrt{N^{-1} \sum_{k=1}^{N} e_k^2}, \quad \mathrm{ATE}_{\max} = \max_k e_k. \tag{108}$$

Each variant uses the same original input. These are descriptive metrics for one sequence, with no independent trial-level confidence interval or statistical significance claim.